

False Alerts Related to GPS Electronic Monitoring in Illinois Cook County Sheriff's Office

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Abstract

The number of people under electronic monitoring has grown due to the increased use of Global Positioning System (GPS) technology. Meanwhile, a rise in false alerts (i.e., violation alerts that turned out to be groundless) from monitoring devices has concerned many as it has strained resources and disrupted lives. Analyzing Illinois Cook County Sheriff's Office electronic monitoring data, we examined whether false alerts increased following the transition to GPS, how false alerts from GPS monitoring were spatially distributed, and to what extent false alerts were associated with GPS drift (position errors). The rate of monthly false alerts increased from 57.79% in 2017–2019 (radio frequency) to 95.50% in 2021–2023 (GPS). Zip codes with high rates of false alerts have tall buildings that are known to cause GPS drift. Analyzing GPS locations of monitored participants in Geographic Information Systems, we found that false inclusion zone violation alert incidents are closely linked to GPS drift. Results indicate that limitations of GPS technology are inadequately addressed in the current arrangement of electronic monitoring involving law enforcement personnel, and private contractors. Sociotechnical reliability should be improved for electronic monitoring to be utilized as a genuinely cost effective and less restrictive alternative to incarceration.

Keywords

electronic monitoring, GPS, false alerts, GIS, Cook County Sheriff's Office

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Backgrounds

Electronic Monitoring Definition and Administration

In the criminal justice system, electronic monitoring is a form of remote surveillance that is used to supervise defendants and offenders in the community at the pretrial and postconviction stage. It tracks a person's whereabouts using devices attached to their bodies or their confined residential locations. Pretrial monitoring is a condition of release from jail for those who are accused but not yet convicted of a crime, to ensure defendants appear in court. Postconviction (or post-trial) monitoring is often employed as a requirement for those sentenced to probation and a requirement for parole for people released from prison, to ensure they comply with court-ordered conditions including refraining from reoffending (Bales et al., 2010). Policy changes to reduce prison overcrowding and cut costs associated with operating jails and prisons in the United States have led to the expansion of electronic monitoring (Brancale et al., 2018; Starr & Cohen, 2021).

From 2005 to 2021, the number of people under electronic monitoring in the United States increased from around 53,000 to more than 250,000 (Pew, 2016; Zhang et al., 2024). This growth was partly driven by a sharp increase in the use of Global Positioning System (GPS) technology. "From 2005 to 2015, use of GPS devices increased thirtyfold, from 2,897 to 88,172, while the use of radio frequency devices declined 25 percent, from 50,132 to 37,706" (Zhang et al., 2024, p. 17). In addition to ankle devices, a growing number of U.S. agencies adopt electronic monitoring apps in smartphones equipped with GPS tracking capabilities. U.S. Immigration and Customs Enforcement (ICE) monitors over 90% of its supervisees using a smartphone app (Zhang et al., 2024). The number of immigrants monitored via the BI SmartLINK[®] app, for example, increased from 12,000 in 2018 to 95,000 in 2022 (Bhuiyan, 2022). The total number of people on electronic monitoring across both the civil immigration and criminal justice systems likely increased to nearly half a million in 2022 (Zhang et al., 2024).

The electronic monitoring device generates an alert based on GPS readings when a supervisee exits an inclusion zone (i.e., places where the supervisee is permitted or required to be) or enters an exclusion zone (i.e., places where the supervisee is prohibited). The device also triggers an alert if it detects tampering (Agnew, 2022; Belur et al., 2020; Brancale et al., 2018). Often, recorded GPS locations can be dozens of meters away from actual locations, especially when GPS signals interfere with built structures. Such inaccuracies, also known as GPS drift, can result in false alerts that apparently signify the violation of electronic monitoring conditions (e.g., inclusion zone violation) but do not correspond to an actual violation. In the U.S. criminal justice system, the task of supervising the hundreds of thousands of people—including responding to alerts—in electronic monitoring programs falls to law enforcement personnel and private contractors. The administration of electronic monitoring in the United States is decentralized across jurisdictional authorities (Zhang et al., 2024). Jurisdictions often outsource some supervision duties to private companies. The electronic monitoring market, with government agencies as the leading end users, is a highly privatized sector with a lack of oversight. And 99% of the electronic monitoring market in North America was dominated by 11 companies—including BI Incorporated (GEO Group) and Track Group—in 2015 (Pew, 2016). There is no comprehensive national reporting requirement for electronic monitoring use in the United States. Some states, like Illinois under the Pretrial Fairness Act—which is a part of SAFE-T Act—are just beginning to implement such requirements at the local level.

Why Electronic Monitoring Programs are Implemented

Proponents of electronic monitoring programs claim that electronic monitoring increases the risks of getting caught, and enhances incentives for compliance (Belur et al., 2020). A 2006 study

observed 75,661 offenders enrolled in back-end home confinement programs in Florida: electronic monitoring was shown to reduce the likelihood of absconding and revocation for a new offense within 2 years after placement (Padgett et al., 2006). According to the meta-analysis of 18 studies that examined one pretrial, and 17 post-trial programs, electronic monitoring was found to extend the time of re-offending (Belur et al., 2020). No significant difference was found in court appearance rates between defendants with and without electronic monitoring in Cook County, Illinois (Agnew, 2022). It is challenging to draw definitive conclusions as to whether electronic monitoring has desired outcomes mainly due to different methodologies employed and inadequate empirical research. Nonetheless, electronic monitoring placed on high-risk offenders appears to be effective (e.g., reduced recidivism and increased safety) at least in the short-term while being generally ineffective for low-risk populations (Belur et al., 2020; Brancale et al., 2018).

Electronic monitoring is generally seen as less punitive than prison sentences by supervisees. Electronic monitoring can give offenders more freedom of movement compared to prison incarceration or house arrest. Electronic monitoring participants with children and stable families preferred electronic monitoring because it allowed them to continue daily life with their loved ones (Mendonca et al., 2018). On offenders who were under GPS monitoring and not completely restricted to their home residence, electronic monitoring did not show negative psychological impacts (Hawkes et al., 2023). Still, electronic monitoring has varied effects on participants' quality of life. Participants experience electronic monitoring as a punitive sanction, citing deprived autonomy, social stigma, and financial burden due to the restricted movement (Chapman et al., 2022; Richter et al., 2021). Kilgore (2022)—who was placed on electronic monitoring for a year—situates electronic monitoring between the U.S. mass incarceration and surveillance capitalism, arguing that it extends carceral control and replaces human judgment with digital surveillance.

Another major theme in the defense of electronic monitoring is that it is considered cheaper than physical incarceration. According to the Administrative Office of the U.S. Courts in 2020, the daily cost of physical incarceration per person was \$108.00, while the daily cost of GPS electronic monitoring was \$4.15 (Starr & Cohen, 2021). The monitoring cost reported above reflects vendor costs and does not consider administrative overhead including risk assessment and compliance checks. Further, equipment and supervision fees are often shifted to the monitored individual. The Cook County Sheriff's Office electronic monitoring program was previously user-funded, meaning that the person being surveilled paid for the program at least partially.

Challenges With Implementing Electronic Monitoring

It is widely noted that agencies running electronic monitoring programs are faced with substantive challenges. Those challenges are primarily concerned with poorly managed technical deficiencies, inadequate staff education and capacity despite complex protocol, and lack of consistent protocol due to the decentralized and unregulated environment as discussed above.

Monitoring devices do not necessarily function as expected due to inaccurate or missing GPS signals, low batteries, and other technical flaws. GPS electronic monitoring devices receive signals from GPS satellites to determine the location of an individual monitored. GPS signals often interfere with or deflect off built structures such as buildings and tunnels, causing signals to be lost or degraded when monitoring devices are indoors or in densely built-up urban areas (Hwang et al., 2018). This results in "GPS drift," a phenomenon where the location recorded by GPS deviates from the actual location. Cellular networks are used to fill the gaps where GPS signals are weak or unavailable, but resulting positioning is significantly less accurate than GPS. The geographic coverage of positioning is reduced in rural areas because cellular

networks are unavailable in proximity (Zhang et al., 2024). Equipment malfunction, signal loss, and inadequate broadband capacity have been identified as barriers to successfully implementing electronic monitoring programs across the studies (Belur et al., 2020). Despite these technological challenges, some see electronic monitoring as a consistently accurate resource: A Cook County judge claimed that “GPS is the most reliable thing in the world” (Chapman et al., 2022).

Substantial training is necessary for staff to be able to administer electronic monitoring programs competently. As discussed above, electronic monitoring programs are complex and administered in a fragmented fashion even within a jurisdiction, involving courts, law enforcement personnel, and private contractors. Zhang et al. (2024) contacted officials in several small or mid-sized counties across the United States who stated that for them, electronic monitoring “was costly and burdensome to administer” (p. 21). Violation alerts require law enforcement staff to conduct home investigations to determine if the violation is valid (Zhang et al., 2024). Constant monitoring implies that enforcement staff must be available 24 hr a day, which has led to burnout and rapid turnover in some electronic monitoring programs (Belur et al., 2020). Individuals monitored are wrongly blamed for violating electronic monitoring rules due to false alerts, to be detailed below. Within the Cook County Sheriff’s Office law enforcement system, ineffective communication and coordination between entities administering electronic monitoring programs led to an increase in caseloads for individual staff members, unclear monitoring practices, and left participants and their advocates unsure how to be successful on electronic monitoring (Agnew, 2022).

Cook County Sheriff’s Office Electronic Monitoring Program

Cook County, Illinois had the largest electronic monitoring caseload of any unit of local government in the United States in 2020 (Agnew, 2022). Until recently, Cook County had two independent electronic monitoring programs run by different agencies, the Sheriff’s Office and the Office of the Chief Judge. The Cook County Sheriff’s Office electronic monitoring program supervised pretrial defendants for decades—over 3000 defendants in 2021. The Sheriff’s Office divides the administration of the program between four different entities. The Sheriff’s staff oversees formal law enforcement duties such as court liaisons. Three private entities manage other aspects of electronic monitoring administration: Track Group provides electronic monitoring equipment and software, Protocol (a Subsidiary of BI Incorporated) handles the electronic monitoring call center and alert management, and Allpoints performs equipment maintenance and manages inventory (Agnew, 2022). Cook County is set to consolidate the Sheriff’s Office program into the one administered by the Office of the Chief Judge in September 2025 (Agnew, 2024; Cohen, 2025).

Electronic monitoring in Cook County, Illinois has been recently criticized for false alerts (Chapman et al., 2022). Alerts are generated when a monitoring device detects a signal that appears to indicate that the supervisee violates rules for electronic monitoring (e.g., inclusion zone violation and tampering). However, alerts are frequently inaccurate due to technological shortcomings or issues with signal coverage and deflection that are discussed above (Belur et al., 2020; Hwang et al., 2018; Zhang et al., 2024). There were over 167,000 alerts from April 1, 2020, to April 1, 2021, and roughly “52.8% of the dispositions in this dataset were either false alarms or equipment malfunctions” (Agnew, 2022, p. 62). A University of Chicago presentation to the Sheriff’s Office staff “noted that 80% of alerts from the GPS electronic monitoring vendor are ‘false positives’ with 15% of Cook County Sheriff’s Office electronic monitoring participants being flagged with a false positive alert each day” (Agnew, 2022, p. 64). In Cook County, people on electronic monitoring have run into frequent difficulty due to false alerts and signal interference. A Cook County Public Defender said, “we know that we’ve seen alerts when people are actually sitting at their home, abiding by the condition of their release” (Chapman et al., 2022).

In 2020, Cook County transitioned its electronic monitoring program from radio frequency to GPS. By January 2021, the entire program was using GPS technology. The caseload per full-time employee in Cook County nearly doubled from 14 in 2019 to 27 in 2021 (Agnew, 2022). This increase in caseload is attributed to the COVID-19 pandemic, as well as the transition to GPS. In September 2023, the Illinois Pretrial Fairness Act took effect. The act eliminated money bail and established a new pretrial release and detention system. In the new system, judges may rely on electronic monitoring if needed instead of money bail: Whether to place defendants on electronic monitoring is assessed based on flight risk and threat to community. While it is too early to determine the full impact of the act on electronic monitoring statewide yet, the act could lead to the expansion of electronic monitoring—especially in counties where it was previously not an option (Nowicki, 2024). Given the anticipated expansion of GPS electronic monitoring programs, false alerts related to GPS electronic monitoring likely continue to cause confusion and difficulties for parties involved.

Aims of the Study

This article focuses on the problem of false alerts, analyzing Cook County Sheriff's Office data. False alert issues warrant further investigation because they may be emblematic of a larger problem in the criminal justice system alongside rising reliance on technology-driven solutions like electronic monitoring. Widespread false alerts can disrupt lives, strain resources, and create a cycle of mistrust that undermines the purpose of electronic monitoring programs. The constant fear of a false alert can cause mental distress among monitored participants. Participants can be incarcerated despite their compliance with their release conditions—without due processes in some cases. With frequent false alerts, officers could potentially miss genuine violations as they become desensitized to alerts. Staff and officers may not have adequate time to investigate alerts, being inundated with a large volume of alerts. This casts doubts about the reliability of electronic monitoring programs. This, if unresolved, can further erode trust between monitored individuals and supervision officers. While electronic monitoring is touted as a cheaper alternative to physical incarceration, the true cost from poorly managed alerts is not often brought to light.

Despite the rising prevalence of GPS electronic monitoring and potential harm of false alerts, little has been known about how extensive false alerts are, and what might be causing false alerts in GPS electronic monitoring. This is largely due to the lack of data reporting requirements on electronic monitoring programs. Existing studies on false alerts, if any, are mostly based on anecdotal evidence and empirical research analyzing comprehensive data, are scarce. The purpose of the study is to portray the status of false alerts during the period transitioning to GPS technology, and present evidence pertaining to how false inclusion violation alerts are associated with GPS position errors—broadly the technology's flaws that are poorly managed. Findings from the study can inform a strategy to reduce false alerts and improve alert management if electronic monitoring continues to be implemented.

In this article, we seek to address the three following questions:

- Question 1: Did false alerts increase after the transition to GPS?
- Question 2: How is the rate of false alerts spatially distributed throughout Chicago?
- Question 3: Are false alerts associated with incorrect GPS locations?

Question 1 is to gain a baseline understanding of the extent and trends of false alerts. We analyze an alerts dataset in 2017–2023 from the Cook County Sheriff's Office electronic monitoring program, to determine whether the transition to GPS technology led to an increase in false alerts.

For the sake of analysis, false alerts are defined as false positive alerts that turned out to be groundless and are identified as a subset of the alerts dataset above that are determined to be false violation. Next, we explore question 2 by mapping the rate of false alerts by zip code in 2022–2023 from the dataset mentioned above. Zip code-level mapping is also utilized to explore how the rate of false alerts is spatially related to factors that are known to cause GPS drift, such as building height. Finally, for question 3, we examine how false inclusion zone violation alerts coincided with incorrect GPS locations by inspecting the Sheriff Office’s spatial trajectory data that tracks locations of participants every minute. To facilitate analysis, incorrect GPS locations are defined as GPS data points (traces) whose recorded locations unambiguously (e.g., at the other side of a block) deviate from actual locations.

Methods

To examine trends in false alerts, we analyzed the violation alert dataset from the Cook County Sheriff’s Office electronic monitoring program. The dataset was acquired through the Freedom of Information Act. The alert dataset comprises violation alerts that were initiated during the period from January 1, 2017 to November 14, 2023. Alerts are generated when any violation of electronic monitoring rules is suspected. Track Group uses proprietary algorithms based on GPS signal quality and adaptive digital filtering to flag invalid GPS location points that may generate false alerts. Protocol running a 24/7 call center determines if an alert is valid or invalid for prioritization by reaching out to participants (Agnew, 2022). Only actionable and valid alerts are forwarded to officers for disposition. The acquired dataset includes alerts that were forwarded to the Sheriff’s Office for adjudication.

Given the institutional arrangements for administering alerts in Cook County Sheriff’s Office electronic monitoring program, “false alerts” correspond to false positive alerts that are deemed to be valid (by a private vendor) but are determined (by officers) to be in no violation of electronic monitoring rules and conditions in this study. The dataset contains information about the type of alerts (e.g., inclusion zone violation and exclusion zone violation), the time the apparent violation began and ended and whether participants violated electronic monitoring rules (i.e., disposition results by officers). True alerts refer to alerts that are determined to violate electronic monitoring rules. We disaggregated the entire alerts by violation status into each year’s monthly periods to depict trends in relation to the transition to GPS technology that occurred in May to December 2020.

To explore the spatial distribution of false alerts in GPS monitoring, we used the dataset described above that spanned from January 1, 2022 to November 14, 2023. This subset includes zip codes of participants’ approved residence location for each alert. We grouped total and false alerts by zip code to map the rate of false alerts for each zip code in Chicago. Moreover, we explored how building height and tree coverage are spatially related to the rate of false alerts. Building height and tree coverage were chosen because tall buildings and dense tree cover are known to cause GPS drift (Abdi et al., 2022; Weng et al., 2023). Building height information is from the Federal Emergency Management Administration (FEMA)’s USA Structure dataset. This dataset contains structures with a size greater than 450 square feet, including building height as a measurement. Our data on tree coverage comes from the 2021 U.S. Forest Service Tree Canopy Cover dataset of the National Land Cover Database (NLCD). Average building height and tree canopy density by zip code were calculated using spatial overlay in Geographic Information Systems (GIS). In addition to thematic maps to depict spatial distribution of three variables, Spearman’s correlation analysis was conducted to determine whether the rate of false alerts is associated with building height and tree canopy density.

To examine how false alerts are associated with incorrect GPS locations, we analyzed the “spatial trajectory dataset” from six participants enrolled in the GPS electronic monitoring

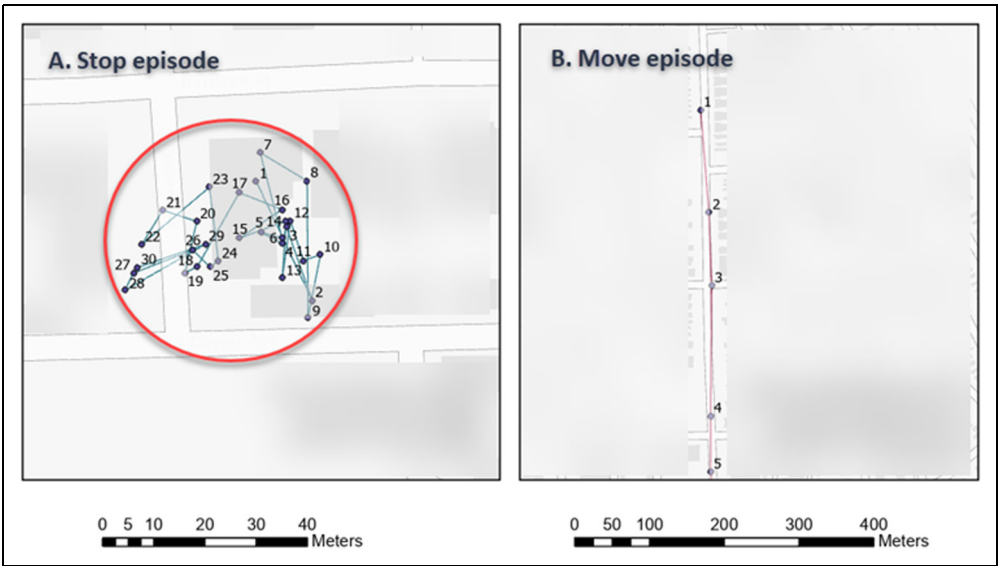


Figure 1. Different error range by motion state for labeling locational outliers.

program by Cook County Sheriff’s Office. Four participants were recruited through a flyer at the People’s Lobby Education Institute within which the Illinois Network for Pretrial Justice is housed. We collaborated with the organization above because the organization, located in the South Side of Chicago, is easily accessible to many electronic monitoring participants, and has a history of working with them. Four recruited participants provided the written informed consent that is approved by Institutional Review Board (protocol # IRB-2023-1040). We also utilized the data from two deceased electronic monitoring participants that Cook County Sheriff’s Office provided. These six participants (four Black and two White) live in different parts of Cook County (2 south side, 2 west side, 1 north side, and 1 suburb), representing the entire population enrolled in the Sheriff’s program in terms of races and residential locations relatively well.

The spatial trajectory data refers to an ordered sequence of locations over time (Zheng & Zhou, 2011). The dataset primarily consists of track points (or traces) that record a participant’s location every minute. The dataset also includes alerts with GPS locations that signal any suspected violation of electronic monitoring rules. All these data points are time-stamped to the second level. Alerts’ data points are generated when suspected violations begin and the apparent violations end (i.e., compliance). From the dataset one can identify a violation alert “incident” spanning from the start time to the end time. We inspected whether any trajectory data points had incorrect GPS locations during the period of each alert incident that may represent either true or false violation.

Three research personnel labeled incorrect GPS locations and the violation status of alert incidents by visually inspecting trajectory data points overlaid on high-resolution (0.3–0.5 m) orthoimages in GIS, in cross-reference to a travel diary (completed by recruited participants to record their location for any trips) and other GPS track logs from QStarz BL-1000st (for recruited participants). Due to substantial difference in error range by movement state, research personnel labeled data differently depending on whether they are stop or move. Figure 1 depicts part of a stop episode that corresponds to a participant staying home for 30 min. In contrast to a move episode (e.g., driving trip), GPS coordinates of a stop episode drift 30 to 40 m from its actual location. If a series of traces fall into a stop episode, the area that traces mostly fall within (a red circle in Figure 1) is regarded as their actual location. Traces outside of their actual location

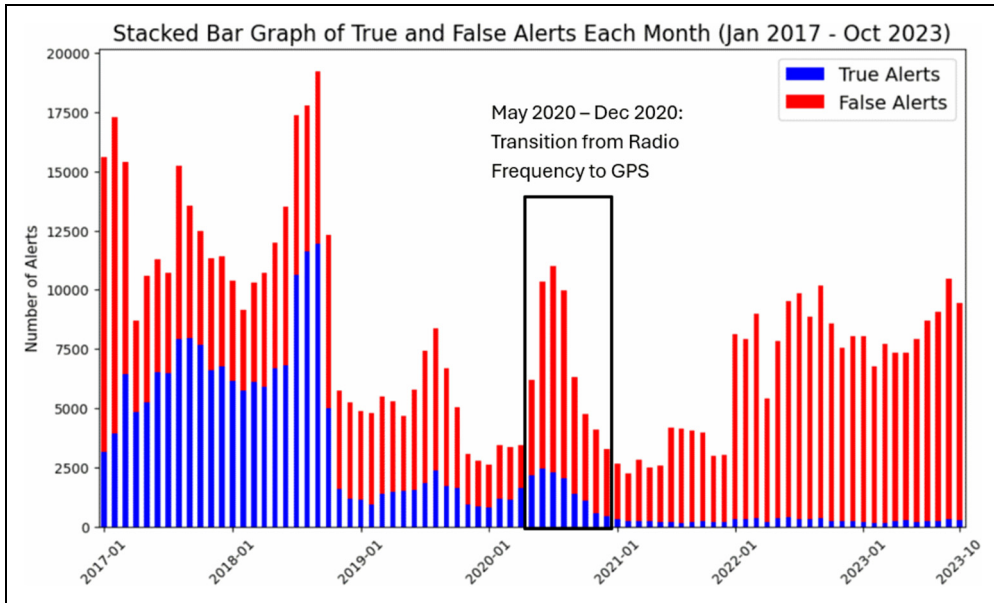


Figure 2. Stacked bar graph of true and false alerts each month from January 2017 to October 2023.

are labeled as incorrect location points. They are typically at least 55 m away from the centroid of their actual location. Fifty-five meters are approximately the sum of average GPS error (5 m) and half of the average block size at the shorter side in Chicago (50 m). If there is disagreement on labeling, research personnel reconcile the differences until an agreement is reached.

If recorded GPS positions during the alert incident resemble a logical journey a person would take (e.g., leaving their residence to drive to a gas station), the alert incident was labeled as a true alert. If recorded GPS positions during the alert incident did not resemble a logical journey (e.g., points scattered across neighborhoods), it was labeled as a false alert. We considered only unambiguously labeled cases for analysis. A Mann–Whitney U test was conducted to determine whether the proportion of incorrect GPS location traces was different between true alert incidents and false alert incidents.

Results

Did False Alerts Increase After the Transition to GPS?

Figure 2 shows the number of monthly alerts by violation status in a stacked bar graph from 2017 to 2023. After the transition from radio frequency to GPS in May to December 2020, the number of false alerts (depicted in red bars) steadily increased. More notably, the rate of false alerts was consistently high after the transition to GPS. Comparing two periods using radio frequency versus GPS entirely, the mean rate of false alerts each month increased from 57.79% (± 15.72 standard deviation) in 2017–2019 to 95.50% (± 2.42) in 2021–2023. During the past 3 years, the rate of false alerts steadily increased: 92.87 (± 2.54) % in 2021, 96.46 (± 0.55) % in 2022, and 97.24 (± 0.56) % in 2023. This analysis provides strong evidence that the number and rate of false alerts increased following the transition to GPS technology. While it is beyond the scope of question 1, it should be noted that the total number of alerts is a function of vendors' filtering methods that have evolved over time. Although GPS monitoring by nature triggers more alerts per person than

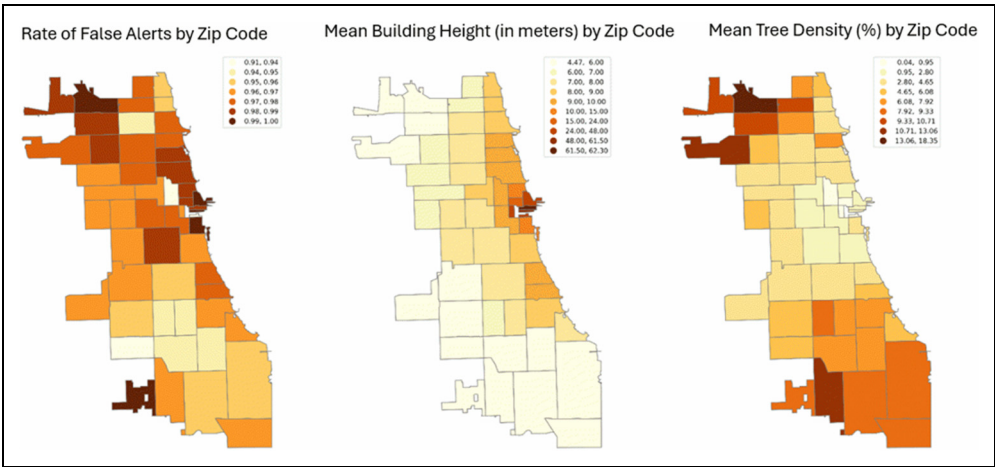


Figure 3. Rate of false alerts, building height, and tree density by zip code in Chicago (2022–2023).

radio frequency that requires participants to be at a fixed location, a smaller number of total alerts was forwarded to officers because vendors furthered the filtering process due to occasional GPS failure (Agnew, 2022).

How is the Rate of False Alerts Spatially Distributed Throughout Chicago?

Figure 3 depicts the spatial distribution of the rate of false alerts, building heights, and tree canopy density by zip code in Chicago. Foremost, rates of false alerts were high, ranging from 91 to 100%, throughout Chicago in 2022–2023. Out of 53 zip codes, 45 zip codes have over 95% false alerts. Two spatial hot spots emerged: areas around the downtown business district (the “Loop”) and the northwestern areas around Forest Glen. Zip codes with 100% false alerts were 60602 (Loop: 4/4), 60646 (Forest Glen: 77/77) with large tree coverage, and 60655 (Mt Greenwood: 7/7) with comparatively higher elevation. In contrast to the rate of false alerts, the building height is highly variable, ranging from 4 to 62 meters, which reflects the concentration of high-rise buildings in a few zip codes around the Loop and near Lake Michigan. The zip codes with the highest average building height are in the Loop, and zip codes farthest from the Loop have the lowest building height. Tree density has an inverse spatial pattern to building height: the farther away a zip code is from the Loop, the more trees there are. Spearman’s rho between the rate of false alerts and building height by zip code ($n=53$) was 0.2226 with a p -value of 0.1092. Spearman’s correlation analysis between the rate of false alerts and tree canopy density provided a rho of 0.153 and a p -value of 0.6047. Although we could not confirm statistical significance in their correlations, Figure 2 could aid researchers in identifying factors influencing false alerts.

Are False Alerts Associated With Incorrect GPS Locations?

For this question, we considered the 77 “inclusion zone violation” alert incidents included in the spatial trajectory dataset. We only examined inclusion zone violations because they appear consistently across multiple participants. Out of six participants, three participants (two recruited and one deceased) had inclusion zone violation alert incidents with a clear end. Table 1 describes labeling results from the 77 alert incidents we examined. Fifty-two of 77 alert incidents were false alerts. Out of 231 traces that make up 52 false alert incidents, 184 traces (79.65%) had incorrect

Table 1. Results of Labeling Incorrect GPS Locations and Violation Status for 77 Inclusion Zone Violation Alert Incidents in the Spatial Trajectory Data.

A. False Alert Incidents						
Type of participants	Tracking period	The number of false alert incidents	The number of false points examined	The number of total track location track points	The number of incorrect GPS location track points	Percent of track points that were incorrect GPS locations
A01 Recruited	01/31/24–02/09/24	2	53	53	53	100
A04 Recruited	07/12/24–07/22/24	4	87	85	85	97.7
B01 Deceased	06/03/20–07/21/20	46	91	46	46	50.55
Total	67 days	52	231	184	184	79.65
B. True Alert Incidents						
Type of participants	Tracking period	The number of true alert incidents	The number of true points examined	The number of total track location track points	The number of incorrect GPS location track points	Percent of track points that were incorrect GPS locations
A01 Recruited	01/31/24–02/09/24	0	0	n/a	n/a	n/a
A04 Recruited	07/12/24–07/22/24	22	1,515	166	166	10.96
B01 Deceased	06/03/20–07/21/20	3	42	0	0	0
Total	25	25	1,557	166	166	10.66

GPS=Global Positioning System.

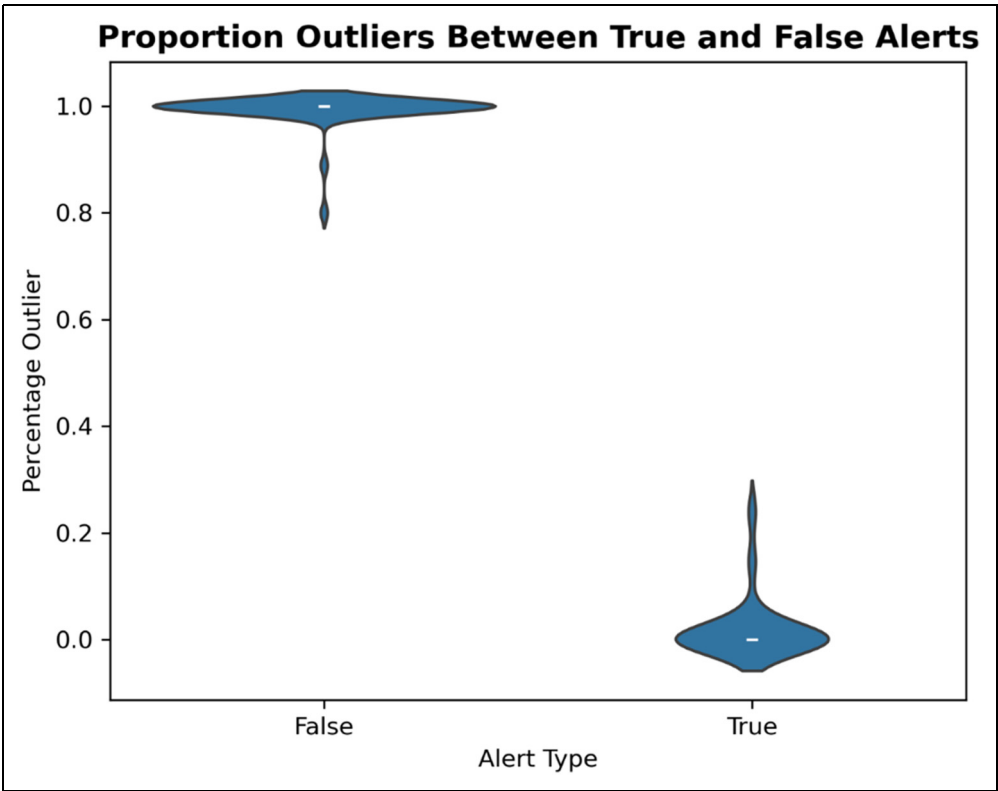


Figure 4. The proportion of locational outliers in 77 alert incidents by violation status in the violin plot. “Outliers” refers to a trace that is an incorrect Global Positioning System (GPS) location.

GPS locations. Twenty-five of 77 alert incidents were true alerts. Out of 1,557 traces making up 25 true alert incidents, 166 traces (10.66%) had incorrect GPS locations.

Figure 4 shows the proportion of locational outliers (traces with incorrect GPS locations) in 77 alert incidents by violation status (false vs. true) in the violin plot. The alerts were mostly binary: Almost all the false alerts were 100% incorrect GPS locations, and almost all true alerts were 0% incorrect GPS locations. Only three false alert incidents (out of 52) contained traces that were accurate GPS locations. Only four true alert incidents (out of 25) contained traces with inaccurate GPS locations. The results of the Mann–Whitney U Test provided a U-value 0.0 with a p -value of 0.000. This result confirms that the proportion of locational outliers in false alert incidents is significantly higher than in true alert incidents. This provides strong evidence that false alerts are associated with incorrect GPS locations.

Discussion

Trends in False Alerts 2017–2023

A clear picture emerged from analyzing the violation alerts dataset from Cook County Sheriff’s Office pretrial electronic monitoring program: there were prevalent “false positive” alerts (453,713 out of 649,477) throughout the analyzed period (January 2017 to October 2023). These alerts apparently indicated violation but were disposed to be no violation of rules by officers. Moreover, the number and rate of false alerts have steadily increased after the transition to

GPS technology (2021 and onwards), reaching 97.24% in 2023 when we analyzed monthly alerts over time. Our findings suggest that flaws of GPS monitoring technology, such as GPS drift and false alerts, are inadequately addressed in the current arrangement of electronic monitoring programs that involve multiple parties including the Sheriff's department, and private vendors. These findings should serve as a forewarning for agencies that consider adopting GPS electronic monitoring in the future. With strong evidence of prevalent false alerts, our analysis warrants further empirical studies that examine potential harm of false alerts involving law enforcement personnel, private contractors, and individuals monitored.

We would have conducted more thorough and multifaceted analysis from the alert dataset if the dataset had been consistent throughout the years. The 2022–2023 dataset we received was vastly different from the dataset of previous years. Otherwise, we could have examined, for instance, which group of participants (e.g., by offense type) experiences more false alerts. Cook County Sheriff's Office did not release some of the recent years' dataset we requested, in line with the time the agency's electronic monitoring program was being phased out. In addition, the data the Sheriff's Office shared with the research team had multiple quality issues. There were numerous instances of duplicate data, incompatibility in alert data between periods, and missing data. It was necessary to spend substantial time cleaning data prior to analysis. For any researchers interested in analyzing electronic monitoring data, careful data wrangling plans will be needed to address data quality issues.

Spatial Distribution of False Alerts 2022–2023

We presented exploratory analysis on how the rate of false alerts is spatially distributed in Chicago. Out of 53 zip codes, 50 zip codes have over 94% false alerts in 2022–2023, indicating that the rate of false alerts is high throughout Chicago. Zip codes with a high rate of false alerts (over 98%) appear to coincide with areas with tall buildings or dense tree canopy that are known to cause GPS drift (i.e., GPS position errors). Spearman's correlation analysis, however, could not confirm statistically significant correlations that are suggested by thematic maps.

These results (no statistical significance) may be reflective of low variation in a dependent variable (mostly 94%–100%), the presence of outliers (e.g., zip codes in and around the Loop), and the spatial resolution of zip code that is too coarse to represent variables like building height and tree coverage. Moreover, there are multiple factors that are known to cause GPS drift that is hypothesized to be linked to false alerts. We did not consider a whole array of those factors in our analysis that is primarily intended to facilitate the process of identifying factors related to false alerts. The FEMA Structures dataset, for instance, only measures buildings: It does not include infrastructure like bridges, highway overpasses, or tunnels that may have the potential to cause GPS errors or signal loss.

False Alert Incidents and GPS Position Errors

Analyzing 77 inclusion zone violation alert incidents from the 67-day spatial trajectory data of three individuals enrolled in GPS monitoring, we found that false alert incidents had a significantly higher proportion of GPS position errors than true alert incidents. A Mann–Whitney U test confirms that false alerts are strongly associated with incorrect GPS locations. This suggests that improving the ability to correctly identify GPS position errors in the spatial trajectory data could significantly reduce false alerts (Hwang & Shaik, forthcoming). False alert incidents typically lasted briefly compared to true alert incidents, which might have increased the proportion of locational outliers in false alert incidents. Analysis of alert incidents from more

participants for longer periods is warranted to corroborate the validity of a conclusion drawn from our analysis.

The limited availability of data from Cook County Sheriff's Office restricted the sample size and labeling methods for question 3. We were only able to examine data from participants the research team recruited directly and from deceased participants. Since Cook County did not provide officers' disposition results of alerts for four recruited participants in 2024, the research team determined the validity of alerts based on visual inspection in GIS. To address this limitation, it is necessary to conduct analysis utilizing a collection of raw alerts including disposition results, filtered alerts, and trajectory data. This large-scale analysis will allow researchers to formulate ways to improve alert management.

Conclusion

This article explores whether false alerts in electronic monitoring have increased in transition to GPS and how false alerts are linked to GPS location errors using the dataset provided by the Cook County Sheriff's Office. Results showed that the rate of false alerts increased since the electronic monitoring program transitioned from radio frequency to GPS. We also discovered that incidences of false inclusion zone violation alerts were closely aligned with incorrect locations in GPS monitoring. The findings indicate that false alerts are widespread in GPS monitoring, and that inaccurate GPS location data is a contributing factor.

False alert issues emerge within a sociotechnical system where policies, technological design, vendor business models, and institutional practices interact. A mismatch exists between the policy regime and the technical capacities of the electronic monitoring system. The system inherently produces more false alerts when policies demand precision beyond what GPS can deliver. Our analysis shows that accurate positioning is not guaranteed indoors or in dense urban environments. Errors in GPS-based location data do not only increase false alerts but also introduce uncertainty in how alerts are interpreted and addressed. GPS blind spots—periods in which GPS or cellular signals are unavailable—can result in false negative alerts (i.e., missed alerts) unless private vendors and/or public agencies engage in proactive outreach and verification.

Some vendors are predisposed to design systems that produce a large volume of alerts. Vendors are more incentivized to lower thresholds for triggering alerts than ensure the accuracy of alerts as they seek to avoid the risk of false negatives (i.e., not recognizing genuine violation). These low thresholds in turn work to increase the rate of false positive alerts. More alerts may justify larger staffing requirements and make large-sum contracts look worthwhile. It is difficult to ensure that the monitoring staff and supervising officers correctly distinguish genuine violations from alerts, overwhelmed by many—mostly low-value—alerts. Staff and officers could respond to alerts differently depending on training and risk culture facing uncertainty of information presented to them.

Addressing false alert issues requires joint efforts by both private vendors and public agencies because the problem is sociotechnical rather than purely technical. Jurisdictions could strengthen oversight by conducting independent audits of electronic monitoring systems and standardizing the reporting of false alert rates. Vendor contract renewals could be tied to performance metrics such as maximum allowable false alert rate and satisfactory alert resolution. Vendors should improve indoor positioning capabilities and location filtering/enhancing algorithms. Private vendors and public agencies can work together to develop standardized alert filtering/triage protocols, drawing from their experiences in the field. In conclusion, sociotechnical reliability of electronic monitoring should be improved for electronic monitoring to be utilized as a genuinely cost effective and less restrictive alternative to incarceration.

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